NANYANG TECHNOLOGICAL UNIVERSITY

SEMESTER 1 EXAMINATION 2024-2025

IE2104 – DIGITAL ELECTRONICS

November / December 2024 Time Allowed: 2 hours

INSTRUCTIONS

1. This paper contains 4 questions and comprises 5 pages.
2. Answer all questions.
3. All questions carry equal marks.
4. This is a closed book examination.
5. Unless specifically stated, all symbols have their usual meanings.

---

1. (a) Draw a CMOS circuit with minimum number of transistors for the logic function $z(a,b,c,d) = c \cdot (\bar{b} + d) + a$.

(7 Marks)

(b) Given $f(a,b,c) = \Pi M(1,3,5,14,15) + x(0,7,9,12)$, where $x(0,7,9,12)$ are don't care terms, find:

(i) the minimized product of sums of $f(a,b,c)$;

(ii) the minimized sum of products of $f(a,b,c)$.

(10 Marks)

(c) People in a remote planet use a radix-r number system. To them, the following equations are both valid. What is the radix r they used?

$$364_r - 64_8 = 45_7 \times 35_r - 369_{16}$$

$$772_r + 320_8 = 6_{16} \times 165_r$$

(8 Marks)

ATTENTION: The Singapore Copyright Act applies to the use of this document. Nanyang Technological University Library

2. (a) Given a logic function $f(a,b,c) = (\bar{c} \oplus a) \cdot (\overline{a \cdot \bar{b}})$, implement the function with the multiplexer of which the function table is given in Table 1. Assume that the complement signals of the inputs are available.

(10 Marks)

Table 1

Select		Output
S_1	S_0	Y
0	0	I_0
0	1	I_1
1	0	I_2
1	1	I_3

(b) Figure 1 shows a logic circuit consisting of a decoder, a multiplexer and two logic gates. The function tables of the multiplexer and the decoder are given in Table 1 above and Table 2 on page 3 respectively. Find the minterms of the logic function of $g(a,b,c,d)$ in the form of $\sum m(\ldots)$.

(10 Marks)

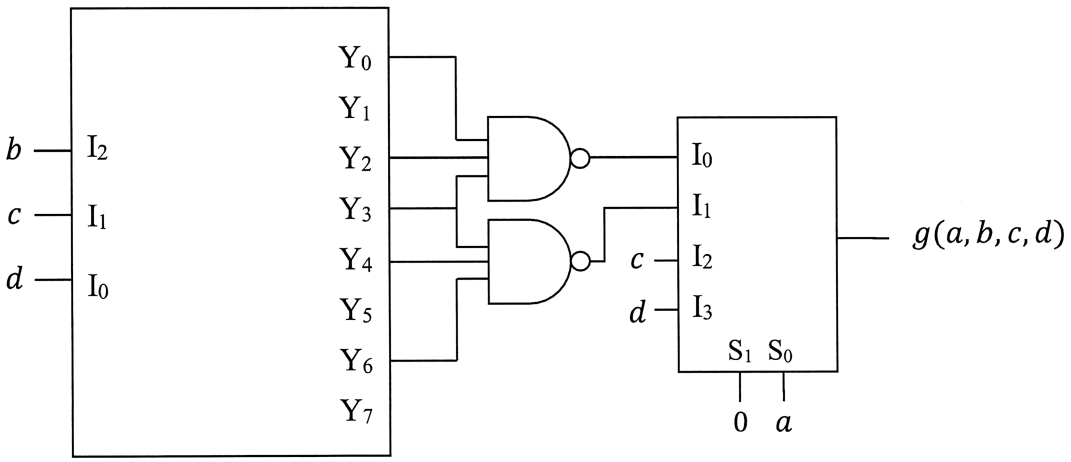


Figure 1

Note: Question No. 2 continues on page 3.

ATTENTION: The Singapore Copyright Act applies to the use of this document. Nanyang Technological University Library

Table 2

Input			Output							
I_2	I_1	I_0	Y_7	Y_6	Y_5	Y_4	Y_3	Y_2	Y_1	Y_0
0	0	0	1	1	1	1	1	1	1	0
0	0	1	1	1	1	1	1	1	0	1
0	1	0	1	1	1	1	1	0	1	1
0	1	1	1	1	1	1	0	1	1	1
1	0	0	1	1	1	0	1	1	1	1
1	0	1	1	1	0	1	1	1	1	1
1	1	0	1	0	1	1	1	1	1	1
1	1	1	0	1	1	1	1	1	1	1

(c) Implement the logic function $t(x, y, z) = \prod M(0, 1, 2, 5, 7)$ with ONLY the logic circuit in Figure 1 on page 2. Use the logic symbol in Figure 2 for the logic circuit and draw the implementation with clear input and output labels.

(5 Marks)

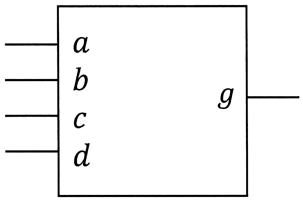


Figure 2

3. (a) Draw a positive level triggered D flip-flop and a positive level T flip-flop based on the cross-coupled NAND latch.

(8 Marks)

Note: Question No. 3 continues on page 4.

ATTENTION: The Singapore Copyright Act applies to the use of this document. Nanyang Technological University Library

(b) A serial adder finite state machine (FSM) is shown in Figure 3. The function of the serial adder FSM is to add input bits a, b and c_{in} in one clock cycle and produce the outputs s and c_{out}. The c_{out} of the previous stage will be used as the c_{in} of the current stage.

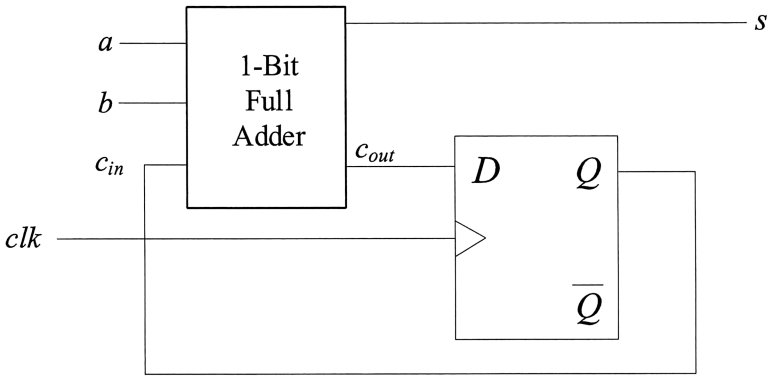


Figure 3

(i) Determine whether the serial adder FSM is a Mealy machine or a Moore machine and state your reasons.

(ii) Write the Boolean expressions for s and c_{out}.

(iii) Tabulate the transition table of the D flip-flop.

(iv) Tabulate the next state and output table.

(v) Draw the state diagram of the FSM.

(14 Marks)

(c) Briefly discuss the possible techniques to reduce the power dissipation of an FSM circuit.

(3 Marks)

ATTENTION: The Singapore Copyright Act applies to the use of this document. Nanyang Technological University Library

4. (a) Design a 4-bit synchronous counter to count the following repeated sequence: 0, 4, 8, 9, 5, 1. If the counter enters into unused states, it should go to state 0 at the next clock cycle. The counter should be designed in the way where 2 T flip-flops are used for the most significant two bits while 2 D flip-flops are used for the least significant two bits. Assume the complement outputs of the flip-flops are available.

(i) Draw the state diagram.

(ii) Tabulate the state transition table.

(iii) Draw the K-maps.

(iv) Write the input equations to the flip-flops

(v) Implement the circuit using only NAND gates.

(19 Marks)

(b) For the counter designed in part (a), it is given that the delay of NAND gate is 7ns, delay of the T flip-flop is 14ns, delay of the D flip-flop is 12ns and the set-up time of all the flip-flop is 8ns. Determine the maximum operating frequency for the counter.

(6 Marks)

END OF PAPER

ATTENTION: The Singapore Copyright Act applies to the use of this document. Nanyang Technological University Library

ATTENTION: The Singapore Copyright Act applies to the use of this document. Nanyang Technological University Library

ATTENTION: The Singapore Copyright Act applies to the use of this document. Nanyang Technological University Library

IE2104 DIGITAL ELECTRONICS

CONFIDENTIAL

Please read the following instructions carefully:

1. **Please do not turn over the question paper until you are told to do so. Disciplinary action may be taken against you if you do so.**

2. You are not allowed to leave the examination hall unless accompanied by an invigilator. You may raise your hand if you need to communicate with the invigilator.

3. Please write your Matriculation Number on the front of the answer book.

4. Please indicate clearly in the answer book (at the appropriate place) if you are continuing the answer to a question elsewhere in the book.

ATTENTION: The Singapore Copyright Act applies to the use of this document. Nanyang Technological University Library